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Hyper-Local Surplus Food Rescue & Safety-Compliant Redistribution Network

🎯 UN SDG 2: Zero Hunger | 🌱 UN SDG 12: Responsible Consumption | 📍 Pilot: UIU Central Cafeteria

👥 PROJECT TEAM ROSTER (5 MEMBERS — ALLOCATED 2 SLIDES EACH / 2 MINS PER MEMBER)

- **Member 1 (Lead & Problem):** [Your Name 1] • ID: [011 2X1 XXX]
- **Member 2 (Evidence & Solution):** [Your Name 2] • ID: [011 2X1 XXX]
- **Member 3 (Core Features):** [Your Name 3] • ID: [011 2X1 XXX]
- **Member 4 (Plan of Action):** [Your Name 4] • ID: [011 2X1 XXX]
- **Member 5 (Timeline & Pilot Defense):** [Your Name 5] • ID: [011 2X1 XXX]

1. Problem Identification: What Problem Are We Solving?

Stating the core problem clearly in simple terms and why it is critically important to solve

THE PLAIN TRUTH

Food Is Dumped While People Sleep Hungry.

In Dhaka City, food waste is **NOT** caused by food shortage.

It is fundamentally an **INFORMATION, COORDINATION, and MICRO-LOGISTICS failure**.

Kitchens prepare in bulk. When trays remain at night, donors lack a rapid, verified channel to reach hungry communities within the critical biological safety window.

01 THE DAILY MIDNIGHT FOOD DUMP

University cafeterias (e.g. UIU Central Cafeteria) and restaurants cook in bulk. At 10 PM, 40–80 kg of edible food is thrown into dumpsters simply because donors lack instant charity contacts.

02 THE 4-HOUR BIOLOGICAL SAFETY CLOCK

Cooked food enters the bacterial danger zone (5°C–60°C). Under BFSA guidelines, it must be eaten within 4 hours. Manual phone calls take 5+ hours, resulting in complete food spoilage.

03 THE MISSING MICRO-LOGISTICS LINK

Local shelters and orphanages have zero standby vehicles, fuel budget, or couriers at night. Food is available 2 km away, but no automated software dispatches riders to deliver it.

💡 **SUMMARY: Food exists and hunger exists within 3 km, but lack an automated software platform to connect them.**

1. Evidence & Data: Backed by Hard Statistics

Authoritative data on national food waste, biological danger limits, and urban food insecurity

10.7M Tons
ANNUAL FOOD WASTE
UNEP Food Waste Index 2024 (BD)

4 Hours
BIOLOGICAL SAFETY LIMIT
BFSA Danger Zone (5°C – 60°C)

3.8 Million
URBAN FOOD-INSECURE
BBS & World Bank Slum Report

0% Digital
CAMPUS TRACEABILITY TODAY
Zero Audit Trail or Safety Records

01

SOCIAL IMPACT: ZERO HUNGER (SDG 2)

Directly feeds destitute children, orphanages, and street laborers with preserved human dignity. Rescuing a single 40 kg batch feeds **80–100 hungry people** every night.

02

HEALTH & SAFETY: BFSA COMPLIANCE

Enforces mandatory digital timestamping and temperature monitoring. Food exceeding the 4-hour threshold is automatically locked out, completely eliminating food poisoning risks.

03

ENVIRONMENT: METHANE PREVENTION (SDG 12)

Diverts tons of decomposing organic food from open city landfills in Aminbazar and Matuail, slashing lethal greenhouse gas emissions and urban environmental decay.

2. Solution: The Core Idea & 3-Sided Ecosystem

An idea-driven, synchronized hyper-local network engineered for 15–30 minute rescue turnaround

☀️ ONE SINGLE LINE (EXPLAINING THE PROJECT TO A KID OR PARENT):

"An Uber-like platform where university cafeterias post extra hot food, and student volunteers bike it to nearby orphanages in 30 minutes before it spoils!"

01

FOOD DONORS

Cafeterias, Caterers & Restaurants

- **60-Second Posting:** Quick 3-field listing form.
- **AI Spoilage Gauge:** Visual countdown dial.
- **Meal Converter:** '40 kg Biryani = 80 meals'.

02

VERIFIED NGOS

Orphanages & Local Shelters

- **Proximity Radar Map:** Filtered to 3-5 km.
- **1-Click Claim Engine:** Instant reservation.
- **Digital CSR Receipts:** Automated audit reports.

03

STUDENT VOLUNTEERS

Campus Cyclists & Bikers Squad

- **Uber-Style Push Alerts:** Notified on commute.
- **15–30 Min Velocity:** Rapid drop-off.
- **Thermal Bag Check:** Insulated food safety.

2. Main Features: 60-Sec Post & Proximity Radar Map

What will the end system be able to perform: Rapid listing, AI spoilage detection, and geo-filtering

FEATURE 01

60-Sec Listing & AI Spoilage Meter

• **What it does:** Cafeteria staff simply enters dish name, quantity, and cooking timestamp in 3 clicks.

• **AI Spoilage Gauge:** Computes real-time biological countdown based on ambient temperature and BFSA safe holding rules.

• **Automated Meal Calculator:** Automatically converts raw kilograms into child servings (e.g., '40 kg Biryani = 80-100 meals').

• **Safety Timeout Lock:** Automatically de-lists food if remaining pickup window drops below 45 minutes to guarantee safety.

FEATURE 02

3-5 km Hyper-Local Radar Map

• **What it does:** Interactive GIS radar map displaying available surplus batches in real-time.

• **Traffic-Proof Proximity:** Strictly prioritizes donations within a 3-5 km radius to bypass Dhaka's gridlock.

• **1-Click Rescue Claim:** Verified NGO centers can lock and claim a food batch with a single tap.

• **Urgency Rings:** Food lots with under 2 hours remaining flash with pulsating red rings as 'Critical Urgency'.

OUTPUT: A donor posts surplus in under 60 seconds; nearby charities see it on the radar map instantly.

2. Main Features: Uber-Style Dispatch & OTP Handshake

What will the end system be able to perform: Crowdsourced student logistics and zero-trust verification

FEATURE 03

Uber-Style Volunteer Dispatch

• **What it does:** Micro-logistics dispatch engine powered by university cyclists and bikers.

• **Smart Commute Alerts:** Broadcasts push notifications to students heading home along the delivery corridor.

• **Turn-by-Turn Navigation:** Displays donor location, recipient shelter, and recommended cycling route.

• **Thermal Carrier Compliance:** In-app checklist prompts volunteers to verify insulated thermal bag hygiene.

FEATURE 04

4-Digit OTP & QR Handshake

• **What it does:** Cryptographic dual-layer verification at pickup and drop-off.

• **Pickup Verification:** Cafeteria donor only releases food when volunteer enters the secret 4-digit PIN (e.g. '7429').

• **Drop-off Handover:** Shelter representative scans dynamic QR token upon arrival to confirm delivery.

• **Anti-Fraud Guarantee:** 100% eliminates food diversion, black-market resale, or false delivery claims.

🔒 **WORKFLOW: 60s Post ➔ AI Spoilage Verification ➔ 3km Radar Alert ➔ Student Biker Dispatch ➔ OTP Handshake Handover**

3. Plan of Action: High-Level Architecture & Pipeline

End-to-end data flow pipeline connecting donors, business logic, GIS, and couriers without deep code

END-TO-END DATA FLOW PIPELINE

01 DONOR WEB (Next.js 16 SSR) → 02 API LOGIC (TypeScript / Zod) → 03 DATABASE (Prisma + PostgreSQL) → 04 RADAR GIS (Leaflet + OSM) → 05 VOLUNTEER PWA (WebSockets & FCM) → 06 OTP HANDOVER (Tamper-Proof PIN)

01
PRESENTATION LAYER

Next.js 16 App Router & PWA

- Responsive Web Interface for Cafeteria Donors.
- Progressive Web App (PWA) for Volunteer Cyclists.
- Server-Side Rendering (SSR) for sub-100ms response.
- Tailwind CSS v4 design system with mobile-first UI.

02
APPLICATION & LOGIC TIER

Node.js Server Actions & Zod

- Real-time AI Spoilage dynamic countdown engine.
- Cryptographic 4-Digit OTP generation & verification.
- Role-based access control (Donor, NGO, Volunteer).
- WebSockets push alert broker for instant courier matching.

03
DATA & GEOSPATIAL TIER

PostgreSQL + Prisma ORM + Leaflet

- ACID relational schema for Listings, Claims & Audits.
- Atomic claim locking to eliminate race conditions.
- OpenStreetMap client-side Haversine distance math.
- Zero-cost geospatial stack without paid API keys.

3. Plan of Action: Technology Selection & Rationale

What technologies will be used and why: Clear justification for frameworks and libraries

01

FRONTEND FRAMEWORK

Next.js 16 + React 19 + Tailwind CSS

What It Does: Unified web portal and mobile PWA for Donors, NGOs, and Volunteers.

Why Chosen:

- Sub-100ms load times over 4G cellular connections.
- Progressive Web App (PWA) eliminates app-store install barrier for student cyclists.
- Tailwind CSS v4 provides lightweight, responsive UI.

02

BACKEND & ORM

Prisma ORM + PostgreSQL + TypeScript

What It Does: Handles data persistence, auth & transactional claim state machine.

Why Chosen:

- ACID transactions prevent race conditions (two NGOs cannot double-claim the same batch).
- End-to-end type safety eliminates runtime schema errors.
- Relational integrity links Donors, NGOs, and Couriers.

03

GEOSPATIAL & MAPS

Leaflet.js + OpenStreetMap (OSM)

What It Does: Renders interactive radar maps, circular radius queries & distance.

Why Chosen:

- 100% open-source without Google Maps billing or credit card dependencies.
- Lightweight client-side Haversine circular radius queries.
- Low battery and memory consumption on mobile devices.

💡 **RATIONALE SUMMARY: Zero recurring cloud fees, strict type-safety, and instantaneous mobile performance.**

4. Project Timeline: 16-Week Implementation Roadmap

Structured semester progression from system architecture to live UIU campus field pilot and defense

01

PHASE 1: ARCHITECTURE

WEEKS 1 – 4

- System Requirements (SRS)
- Prisma relational schema models
- 3-Portal responsive UI setup
- Role-based authentication

Status: 100% Completed

02

PHASE 2: CORE LOGIC

WEEKS 5 – 8

- 60s listing form & meal calculator
- AI Spoilage dynamic countdown
- Leaflet geospatial radar (3km)
- 4-Digit OTP & QR handshake

Status: 90% Completed

03

PHASE 3: CAMPUS PILOT

WEEKS 9 – 12

- UIU Cafeteria live trial run
- Student cyclist squad onboarding
- Satarkul orphanage delivery test
- Turnaround latency tuning

Target: Live Field Run

04

PHASE 4: AUDIT & DEFENSE

WEEKS 13 – 16

- Concurrency & stress testing
- Anti-fraud security penetration test
- Project impact analytics report
- Final SE Lab defense presentation

Target: Final Defense

🎯 **MILESTONE OBJECTIVE: Production-grade delivery on schedule with a verified campus field pilot.**

Target Pilot, Project Summary & Q&A Session

Transforming campus surplus into immediate hunger relief with rigorous software engineering

📍 FIELD VALIDATION

UIU Campus Pilot Commitment

- **Primary Donor Site:** UIU Central Cafeteria (closing hours surplus).
- **Courier Squad:** UIU student cyclists commuting toward Satarkul & Badda.
- **Recipient Beneficiaries:** Local Madrashas, Orphanages & Shelter Homes within 3 km.
- **Evaluation Metric:** Measuring average rescue latency (<30 mins) and food temperature compliance (>65°C).

🚀 PROJECT SUMMARY

Zero Waste • Zero Hunger • Safe Delivery

- **Idea:** Turn 40–80kg daily campus food waste into 80–100 meals.
- **Technology:** Next.js 16 + Prisma + Leaflet GIS with zero operating cost.
- **Integrity:** 4-digit OTP & QR protocol prevents food theft and fraud.
- **Academic Rigor:** Complete adherence to SE Lab software development standards.

THANK YOU, RESPECTED FACULTY & PEERS! • OPEN FOR QUESTIONS & FEEDBACK